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4th Networking International Conference on Emerging Trends in Expert Systems, Applications & Security, ASIA : Submission (116) has been edited.

1 message

Microsoft CMT <noreply@msr-cmt.org>
To: rounakbharti2004@gmail.com

Tue, Sep 15, 2026 at 6:50 PM

Hello,

The following submission has been edited.

Track Name: NICETASASIA2026

Paper ID: 116

Paper Title: Explainable Risk Assessment Across Code and Synthetic Media: A Hybrid Rule-Learning Framework

Abstract:

We introduce two problems: detecting vulnerabilities in source code, and detecting deepfakes in images, with SecurAI Analyzer being a hybrid solution, comprising a set of rules and machine learning models. In most papers in this area, the decisions on the design of their fusion, the amount of training data they use, and the benchmark setup are made in advance and the various decisions are seldom tested. All three were tested, however. The combination of 37-rule engine and a Random Forest Classifier is the basis for Case Study 1. Only a problem with the original training set was that we found and fixed, then wrote an alternative held out benchmark of 50 cases. This is where the engine gets a complete recall and 100% precision (also providing Wilson confidence intervals on this). We then tested what weight the score should be given to the classifier (0 doesn't affect the precision which drops to about 70% when we give it weight

1) and decided to leave its weight zero, and only present it as additional evidence. Case Study 2 uses an older, custom-trained CNN, but also six classic forensic checks, and a public ViT-based model, which was trained on 2.7 million images from 4,803 different generators. We tested the two at the same time: the first model alone achieves 0.9758 AUC and the second combined with the forensic signal achieves 0.9678 AUC - we report this difference with a sufficiently rough confidence interval, rather than assuming it is a sure thing. Therefore the forensic signal will also score as 0 points when being scored but is a piece of evidence to be displayed. We compare these results to 15 recent papers in the two research areas, show an architecture diagram and an algorithm comparison, and are honest about limitations and testing design decisions: we feel that it is a valuable contribution.

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Primary Subject Area: Multimedia Applications in Forensics, Security & Intelligence

Secondary Subject Areas: Not Entered

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